

BUS 301:

Theory:

From Mid:

1) Question: Explain "Company is an artificial person."

Answer:

A company is recognized as an artificial or juristic person under law. This means that although a company is not a natural human being, it enjoys many legal rights and obligations similar to a natural person.

A company can:

- **Be sued or sue others in court**
- **Enter into contracts and agreements**
- **Open bank accounts**
- **Own property**
- **Conduct business in its own name**

The concept of artificial personality gives the company a separate legal identity distinct from its members.

2) What is partnership business? Explain the type of Partners.

A partnership business is a business between 2 to 20 persons with agreed, written or registered agreement called a deed.

Types of partners:

1) Nominal partners: Lend their name to the partnership but do not actively participate or invest capital. They remain liable for partnership debts.

2) Sleeping partners: Contribute capital but do not participate in day-to-day management. They share profits and losses but remain dormant.

3) Ordinary partners: Active partners who participate in management and operations. They invest capital, make decisions, and share profits and losses.

4) Limited partners: Have limited liability restricted to their capital contribution. They do not participate in management.

5) Partners by holding out: Represent themselves as partners even though they may not be actual partners. They become liable to third parties who rely on such representation.

3) Explain the steps of forming a private limited company.

Ans: These are the necessary steps of forming a private limited company:

Step 1: Name Clearance from RJSC

Step 2: Drafting the Company Documents

Step 3: Opening a Bank Account (for Foreign Shareholders)

Step 4: Submitting Documents to RJSC

Step 5: Obtaining the Certificate of Incorporation

Step 6: Post-Incorporation Formalities

Step 7: Compliance and Operational Readiness

4) Difference between industry and commerce.

Industry: The means of production - the process of collecting natural resources, converting it to raw material and through processing the raw materials, finished goods are prepared for uses.

Commerce: The exchange of goods and services between individuals, businesses, or nations, along with all activities that facilitate this exchange including trade and auxiliary services like banking, transportation, insurance, warehousing, and advertising.

5) A business idea starts with the proper analysis of needs, wants, and demands." Explain this statement with examples.

Ans:

In marketing, a market refers to a place or system where buyers and sellers interact to exchange goods and services, either physically or digitally. It consists of people with needs or wants, the ability to buy, and the willingness to buy.

Needs: Basic human requirements for survival and well-being. Such as, Food, water, shelter, clothing, safety, education, healthcare. It is universal and inherent; not created by society or marketing.

Wants: The form human needs take as shaped by culture, society, and individual personality. A need for food may become a want for pizza, sushi, or biryani. Not essential for survival but are shaped by personal preferences and social influences.

Demands: Wants backed by the ability and willingness to pay. For example, a person may want an iPhone, but it becomes a demand only if they can afford and are willing to buy it.

Final Theory:

1) Writing a business plan. (Scenario based Templates)

এখানে ৩ টা সিনারিও বানানো আছে, ঐ অনুযায়ী Analysis and Planning করা আছে।

আইডিয়া নিয়ে প্রশ্নানুযায়ী লিখতে হবে, নিচের টপিক গুলো Mention করে।

A,B section: নিচের sample answer এ যা আছে, এইগুলো লিখলেই হবে।

C,D section: 12 ta aspect mention kore likhte hobe. Nicher pic e topic dewa ache. (Thanks Anamika)

[This pic is only for C,D section people]

Business Plan Outline	
Cover Page	• Business name, logo, tagline • Contact info (address, email, website, phone) • Date
Table of Contents	
I. Executive Summary	• Problem Identification (Problem Statement) • Solution • Market Gap • Quick financial highlights • Business Objectives
II. Industry Analysis	• Industry trend (Porter's Five Forces Framework, PEST/ PESTLE analysis) • Industry size, growth rate, projections • Key success factors • Emerging industry trends (AI, sustainability, digital adoption)
III. Market Analysis	• Problem Identification • Solution • Market gap & opportunity mapping • Target market & customer segments • Buyer personas and behavior insights • Competitor analysis (with SWOT matrix) • Competitive Differentiation (what sets you apart — price, quality, innovation, service)
IV. Company Description	• History and background • Mission & vision statements • Products/services overview • Legal structure and ownership • Taxation & Compliance Requirements
V. Business & Revenue Model (Products and Services)	• Detailed Description of Products/Services • How the business makes money (subscription, one-time sales, ads, freemium, etc.) • Pricing strategy • Product/Service Lifecycle
VI. The Economics of the Business and Financial Plan	• Startup Costs & Capital Requirement • Cost structure (fixed & variable) • Break-even Analysis • Projected Income Statement • Balance Sheet Forecast
	• Cash Flow Forecast • Financial Ratios & KPIs
VII. Funding Request	• Determining the source of funds (venture capital, angel fund) • Use of Funds (working capital, expansion, marketing, etc.) • Capital structure • Investor Return/Exit Strategy
VIII. Marketing & Growth Plan	• Marketing Mix (4P) • Digital marketing (SEO, content, paid ads, social media, influencer strategy) • Offline marketing (events, PR, networking) • Customer acquisition funnel • Retention strategy (loyalty programs, CRM)
IX. Design & Development Plan	• Development roadmap (MVP → Beta → Launch → Scaling) • Technology stack / R&D efforts • Challenges, risks & mitigation strategies • Intellectual property strategy (patents, trademarks, copyrights)
IX. Operations Plan	• Day-to-day operations workflow • Supply chain strategy (if applicable) • Business location(s) • Facilities & equipment • Quality assurance & compliance
X. Management & Organizational Structure	• Org chart • Organizational Structure • Advisors, mentors, and board members • Staffing Plan & HR Policies • Company culture & values
XI. Risk Analysis & Mitigation	• Internal & external risks (market, financial, operational, legal) • Contingency planning • Insurance & compliance measures
Appendix	

Scenerio 1:

COMPLETE BUSINESS PLAN - SCENARIO BASED

Scenario: TechEdu Solutions - Online Learning Platform for Professional Skills

1. COMPANY ANALYSIS

Company Overview & Background: TechEdu Solutions is an ed-tech startup founded in 2025 providing professional skill courses for working professionals and graduates in Bangladesh, with experienced founding team having 15+ years combined expertise.

Vision, Mission & Core Values: Vision: Bangladesh's leading practical skill platform by 2030. Mission: Empower 10,000+ learners with industry-ready skills. Core values: accessibility, quality education, practical outcomes.

Competitive Advantages: Experienced management, learn-by-building methodology, industry partnerships for real projects, personalized learning paths with job placement assistance.

2. INDUSTRY ANALYSIS

Industry Overview: Online education market size BDT 500 crore, 65% internet penetration, driven by smartphone adoption and demand for skill-based education.

Main Players: 10 Minute School (35% market share), Bohubrihi (programming), Brain Station 23 Academy (corporate), Coursera/Udemy (international platforms).

Industry Size, Growth Rate & Profitability: 25% annual growth, 20-30% net margins, low marginal costs, high customer lifetime value.

Five Forces Model:

- **Threat of New Entrants:** Moderate - low barriers to entry but high brand building costs
- **Threat of Substitutes:** High - YouTube, free resources, traditional institutions
- **Rivalry Among Existing Firms:** High - 10 Minute School, Bohubrihi, international platforms competing

- **Bargaining Power of Suppliers: Low** - abundant content creators and mentors available
 - **Bargaining Power of Buyers: Moderate** - price-sensitive market but high switching costs after enrollment
-

3. MARKET ANALYSIS

Target Market: 25-35 year working professionals (60% revenue) and 21-25 year fresh graduates (40% revenue) in Dhaka, Chittagong, Sylhet.

Buyer Persona: "Rifat" - 28-year-old developer, BDT 45,000/month, needs cloud/DevOps skills, prefers practical project-based learning.

Competition Analysis: Bohubrihi (programming), Brain Station 23 (corporate), Datacamp (data science). Our edge: personalized paths, mentors, job placement, real project portfolios.

4. PRODUCT INTRODUCTION

Product Description: Online courses (web dev, data science, digital marketing, UI/UX) with videos, live sessions, projects, certification, coding environments, mentors.

USP: "Learn by Building Real Products with Industry Mentors" - students work on client projects, building portfolios. Based on research: 73% struggle with practical application.

Problem it Solves: Skill-job mismatch, lack of practical experience, expensive bootcamps. Flexible learning while maintaining jobs.

Motivation: Address hiring challenges - 68% tech companies face talent shortage. Goal: 10,000 employment opportunities by 2028.

5. MARKETING PLAN (4P ANALYSIS)

Product: Three tiers - Basic (BDT 15,000), Pro (BDT 25,000), Premium (BDT 40,000). 3-6 months, modular, lifetime access.

Price: Penetration pricing, 30% launch discount, EMI via bKash/Nagad, 20-30% lower than international platforms.

Place/Distribution: Website, mobile app, B2B university/corporate partnerships, physical centers in divisional cities, cloud infrastructure.

Promotion (Direct Marketing): Email to 50,000+ database, SMS notifications, WhatsApp groups, personalized sales calls for Premium.

Promotion (Indirect Marketing): Facebook/Instagram ads (22-35 age), YouTube tutorials, SEO, influencer partnerships, PR in tech blogs. Budget: 40% digital ads, 30% content, 20% partnerships, 10% traditional.

6. ECONOMICS OF BUSINESS

Revenue Drivers: Course sales (Basic, Pro, Premium tiers). Average price BDT 25,000-29,000 per student across years 2026-2030.

Gross Margin: Selling price minus cost of goods sold (content creation, mentor costs). Gross margin improves from 60% (2026) to 69.4% (2030) as content costs amortize over larger student base.

Fixed Costs: Rent, permanent staff salaries, insurance, depreciation. Remains constant: office rent BDT 2L/year, core team salaries, platform maintenance.

Variable Costs: Direct content creation per course, mentor fees per student, marketing per acquisition. Increases with student numbers: materials, instructor payments, student support.

Break-Even Point: Total revenue equals total costs at 1,200 students in Q3 2027. Beyond this point, business becomes profitable.

Pro-Forma Revenue: 2026: 500 students, BDT 125L | 2027: 1,500 students, BDT 390L | 2028: 3,000 students, BDT 810L | 2029: 5,500 students, BDT 1,540L | 2030: 8,000 students, BDT 2,320L

Fund Required: BDT 80L total - Technology 30%, Content 25%, Marketing 25%, Operations 15%, Working Capital 5%. Funding: BDT 60L investors, BDT 20L founders.

END OF BUSINESS PLAN

Scenerio 2:

COMPLETE BUSINESS PLAN - SCENARIO BASED

Scenario: GreenRide - Electric Vehicle Ridesharing Platform

Tagline: "Ride Green, Live Clean"

1. COMPANY ANALYSIS

Company Overview & Background: GreenRide launched in 2025 as Bangladesh's first electric vehicle ridesharing platform, combining eco-friendly transportation with affordable mobility solutions. Founded by ex-Pathao executives with 12+ years combined experience in ride-hailing and sustainable transport sectors.

Vision, Mission & Core Values: Vision: Leading sustainable urban mobility provider in South Asia by 2030. Mission: Reduce carbon emissions by 50,000 tons annually while providing affordable transport. Core values: sustainability, affordability, innovation, customer safety.

Competitive Advantages: First-mover in EV ridesharing, government partnership for charging infrastructure, 30% lower operating costs than petrol vehicles, tech-driven route optimization, trained eco-conscious drivers.

2. INDUSTRY ANALYSIS

Industry Overview: Bangladesh ride-hailing market valued at BDT 3,000 crore, growing 18% annually. Dhaka's 2 million daily ride-hailing trips create massive opportunity. Government's EV policy 2023 offers tax exemptions and subsidies supporting industry growth.

Main Players: Pathao (45% market share), Uber (30%), Obhai (15%), traditional CNG auto-rickshaws (fragmented). No major player in EV segment yet, creating blue ocean opportunity.

Industry Size, Growth Rate & Profitability: Market size BDT 3,000 crore, 18% CAGR projected. Successful players achieve 12-15% net margins. High customer retention (70%) and 3.5 rides per user weekly indicate strong profitability potential.

Five Forces Model:

- **Threat of New Entrants: Moderate** - high capital for EV fleet but government support available
 - **Threat of Substitutes: High** - public transport, CNGs, traditional ride-hailing, personal vehicles
 - **Rivalry Among Existing Firms: High** - Pathao, Uber dominate with aggressive pricing wars
 - **Bargaining Power of Suppliers: Moderate** - limited EV manufacturers (Walton, Runner) but growing options
 - **Bargaining Power of Buyers: High** - price-sensitive market, low switching costs, multiple alternatives
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3. MARKET ANALYSIS

Target Market: Urban professionals aged 25-40 in Dhaka, Chittagong (70% revenue) earning BDT 40,000+/month, environmentally conscious middle-class. Secondary: corporate B2B contracts for employee transportation (30% revenue).

Buyer Persona: "Nadia" - 32-year-old marketing manager, BDT 65,000/month salary, commutes 12 km daily to Gulshan office. Concerns: air pollution affecting her child's health, rising fuel costs, traffic congestion. Values: eco-friendly choices, reliable service, safety.

Competition Analysis: Pathao (largest network, bike/car options), Uber (premium positioning), Obhai (motorbike focus). Our edge: 40% lower fares than competitors, zero emissions appeal, smoother EV ride experience, government backing.

4. PRODUCT INTRODUCTION

Product Description: App-based EV ridesharing with electric cars and auto-rickshaws. Features include real-time tracking, cashless payment, carbon savings tracker, in-app charging station locator, female driver option, corporate account management.

USP: "Save Money, Save Environment - 40% Cheaper, 100% Emission-Free" - leveraging EV's lower operational costs (BDT 2/km vs BDT 6/km petrol). Determined through pilot: customers willing to wait 2 extra minutes for 40% discount and eco-benefit.

Problem it Solves: Air pollution (Dhaka ranks 3rd most polluted city globally), high transportation costs eating 15-20% of middle-class income, unreliable public transport. Provides guilt-free affordable mobility aligned with growing environmental consciousness.

Motivation: Founders' personal experience with respiratory issues from pollution. Bangladesh loses BDT 52,000 crore annually from air pollution health impacts. Goal: prevent 50,000 tons CO2 emissions, save customers BDT 500 crore in transport costs by 2029.

5. MARKETING PLAN (4P ANALYSIS)

Product: Two services - GreenRide Auto (electric rickshaw, 5km range) and GreenRide Car (electric sedan, 15km+ range). Subscription plans: Daily (BDT 200 unlimited), Monthly (BDT 4,000), Pay-per-ride. Features: AC cars, phone charging, WiFi.

Price: Penetration pricing - BDT 12/km (40% below market). Launch offer: first 10 rides at 50% off. Corporate bulk discounts 25%. Dynamic pricing during peak hours (20% surge). Student discounts 15% with verification.

Place/Distribution: Mobile app (iOS/Android) primary channel. Partnership with 50 charging stations across Dhaka. B2B sales team for corporate accounts. Service zones: Gulshan, Banani, Dhanmondi, Uttara initially, expanding to 15 zones by 2027.

Promotion (Direct Marketing): SMS campaigns to Pathao/Uber user database (purchased legally), email marketing to corporate HR departments, WhatsApp business for customer service, door-to-door flyer distribution in target neighborhoods, direct sales calls for B2B.

Promotion (Indirect Marketing): Facebook/Instagram ads highlighting cost savings calculator, YouTube videos showing environmental impact, influencer partnerships with eco-activists, PR in newspapers about sustainability, sponsorship of environmental events. Budget: 50% digital, 25% B2B sales, 15% PR, 10% traditional media.

6. ECONOMICS OF BUSINESS

Revenue Drivers: Per-ride fare (70% revenue), corporate monthly contracts (20%), in-app advertising (5%), data analytics services to urban planners (5%). Average ride value BDT 80, 8-12 rides per vehicle daily.

Gross Margin: Fare (BDT 80) minus driver commission 35% (BDT 28) minus electricity cost (BDT 4) minus vehicle depreciation per ride (BDT 8) = BDT 40 gross profit (50% margin). Margin improves to 58% by 2029 with scale economies.

Fixed Costs: Office rent (BDT 5L/month), permanent staff salaries (25 employees, BDT 18L/month), charging station lease (BDT 3L/month), insurance (BDT 2L/month), tech platform maintenance (BDT 4L/month). Total fixed: BDT 32L/month = BDT 384L/year.

Variable Costs: Driver commissions (35% of fare), electricity (BDT 2/km), vehicle maintenance (BDT 2,000/vehicle/month), customer acquisition (BDT 150/customer), payment gateway fees (2%), marketing per ride (BDT 5). Scales directly with ride volume.

Break-Even Point: Need 50,000 rides/month at BDT 40 gross profit/ride = BDT 20L/month to cover BDT 32L fixed costs + 12L variable. Achievable with 100 vehicles doing 17 rides/day at 65% capacity utilization. Expected by Month 14 (Q2 2027).

Pro-Forma Revenue: 2026: 25,000 rides/month avg, BDT 240L revenue | 2027: 60,000 rides/month, BDT 720L revenue | 2028: 120,000 rides/month, BDT 1,440L revenue | 2029: 200,000 rides/month, BDT 2,400L revenue | 2030: 300,000 rides/month, BDT 3,600L revenue

Fund Required: BDT 500L total - EV fleet purchase 50% (BDT 250L for 100 vehicles), charging infrastructure 20% (BDT 100L), technology platform 15% (BDT 75L), marketing 10% (BDT 50L), working capital 5% (BDT 25L). Funding: BDT 350L from green venture capital, BDT 100L government EV subsidy, BDT 50L founders.

END OF BUSINESS PLAN

3) COMPLETE BUSINESS PLAN - SCENARIO BASED

Scenerio 3:

Scenario: FreshBox - Smart Refrigerator Food Storage Container

Tagline: "Keep Fresh, Waste Less"

1. COMPANY ANALYSIS

Company Overview & Background: FreshBox launched in 2025 by engineer Karim Ahmed, producing IoT-enabled food storage containers that monitor freshness and reduce food waste. Single product focus with patent-pending smart sensor technology developed over 2 years of R&D.

Vision, Mission & Core Values: Vision: Eliminate household food waste in Bangladesh by 2030. Mission: Save 1 million kg food from waste annually through smart storage solutions. Core values: innovation, sustainability, simplicity, affordability.

Competitive Advantages: Patent-pending freshness sensor technology, 60% lower price than imported smart containers, local manufacturing reducing costs, smartphone app integration, understanding of Bangladeshi kitchen needs and eating habits.

2. INDUSTRY ANALYSIS

Industry Overview: Bangladesh kitchen products market BDT 1,200 crore, growing 15% yearly. Smart home products gaining traction with 40% urban smartphone penetration. Food waste costs average household BDT 3,000/month creating strong problem-solution fit.

Main Players: Tupperware (imported, premium pricing), local plastic container manufacturers (RFL, Partex - no smart features), Chinese imports via Daraz/e-commerce (quality issues). No local smart container manufacturer exists.

Industry Size, Growth Rate & Profitability: Kitchen storage segment BDT 300 crore, 12% annual growth. Profit margins 35-45% for branded products. High repeat purchase rate (customers buy 5-8 containers on average).

Five Forces Model:

- **Threat of New Entrants:** Low - requires technical expertise and patent protection

- **Threat of Substitutes: Moderate** - traditional containers, refrigerator organization methods
 - **Rivalry Among Existing Firms: Low** - no direct smart container competitor in Bangladesh
 - **Bargaining Power of Suppliers: Low** - multiple plastic and electronic component suppliers available
 - **Bargaining Power of Buyers: Moderate** - price-sensitive but value innovation for food savings
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3. MARKET ANALYSIS

Target Market: Urban households in Dhaka, Chittagong with BDT 50,000+ monthly income (primary 70%), restaurants and food businesses (secondary 30%). Focus on health-conscious families aged 30-45 with children.

Buyer Persona: "Sadia" - 35-year-old working mother, household income BDT 80,000/month, cooks daily for family of 4. Frustrated with food spoiling quickly, wasting BDT 4,000/month on spoiled vegetables, meat, dairy. Owns smartphone, uses food delivery apps.

Competition Analysis: Tupperware (BDT 500-1,200/container, no smart features), RFL containers (BDT 80-200, basic), Chinese smart containers on Daraz (BDT 2,000+, unreliable). Our edge: smart features at affordable price, local support, Bangla app, food waste calculator.

4. PRODUCT INTRODUCTION

Product Description: 500ml/1L/2L airtight containers with built-in freshness sensors detecting spoilage through gas emissions. LED indicators (green/yellow/red) show freshness status. Smartphone app sends notifications before food spoils, tracks waste savings, suggests recipes for stored items.

USP: "Smart Sensors Save Your Food and Money - Know Before It Spoils" - determined through 200 household tests showing 65% reduction in food waste. Only smart container made in Bangladesh at 70% lower price than imports.

Problem it Solves: Food spoilage waste (average family wastes 30% purchased food), lack of visibility into refrigerator contents, forgetting stored items, health risks from spoiled food. Saves BDT 2,500/month average per household tested.

Motivation: Founder's personal experience wasting BDT 5,000/month on spoiled food for family. Bangladesh wastes 25% of food production worth BDT 50,000 crore annually. Goal: help 100,000 households reduce food waste by 50% within 3 years.

5. MARKETING PLAN (4P ANALYSIS)

Product: Three sizes - Small (500ml, BDT 350), Medium (1L, BDT 450), Large (2L, BDT 550). Starter kit (3 containers + 1 app subscription) BDT 1,200. Made from food-grade BPA-free plastic, dishwasher safe, replaceable sensors lasting 2 years.

Price: Value pricing - BDT 350-550 per container (70% below imported smart containers, 3x above basic containers). Launch discount 30% first month. Bulk discounts: Buy 5 get 1 free. Restaurant B2B pricing 25% wholesale discount.

Place/Distribution: E-commerce (own website, Daraz, Chaldal) 60% sales, retail partnerships (Swapno, Meena Bazar, Unimart) 30%, direct B2B sales to restaurants 10%. Fulfillment through RedX/Pathao courier. Stock in 50 retail outlets by Year 1.

Promotion (Direct Marketing): Facebook Messenger chatbot for customer queries, email newsletters with food storage tips, WhatsApp groups for customer community, SMS reminders for sensor battery replacement, direct calls to restaurant chains for B2B.

Promotion (Indirect Marketing): Facebook/Instagram ads targeting women 28-45, cooking bloggers/influencer partnerships (10 influencers), YouTube recipe videos featuring product, Google Ads for "food storage" searches, in-store demos at supermarkets. Budget: 45% social media, 30% influencers, 15% in-store, 10% Google Ads.

6. ECONOMICS OF BUSINESS

Revenue Drivers: Container sales (85% revenue at BDT 450 average), app premium subscription BDT 50/month (10%), replacement sensors BDT 100 every 2 years (5%). Average customer buys 6 containers = BDT 2,700 initial purchase.

Gross Margin: Selling price BDT 450 minus production cost (plastic BDT 60, sensor BDT 80, assembly BDT 30, packaging BDT 20) = BDT 190 total cost. Gross profit BDT 260 (58% margin). Improves to 63% by Year 3 with volume production.

Fixed Costs: Factory rent BDT 2L/month, permanent staff salaries (8 people, BDT 3.5L/month), machinery depreciation BDT 1L/month, app maintenance BDT 0.5L/month, utilities BDT 0.8L/month. Total fixed: BDT 7.8L/month = BDT 93.6L/year.

Variable Costs: Raw materials per unit (BDT 190), packaging/shipping per unit (BDT 40), sales commission 10% (BDT 45), marketing per unit (BDT 30), payment gateway 3% (BDT 14). Total variable BDT 319/unit sold.

Break-Even Point: Fixed costs BDT 7.8L/month divided by contribution margin BDT 131/unit (BDT 450 - BDT 319) = 5,954 units/month. Achievable at 200 units/day production capacity with 99% utilization. Expected by Month 9 (Q4 2026).

Pro-Forma Revenue: 2026: 4,000 units/month avg, BDT 216L revenue | 2027: 8,500 units/month, BDT 459L revenue | 2028: 15,000 units/month, BDT 810L revenue | 2029: 22,000 units/month, BDT 1,188L revenue | 2030: 30,000 units/month, BDT 1,620L revenue

Fund Required: BDT 120L total - Manufacturing equipment/molds 35% (BDT 42L), initial inventory raw materials 25% (BDT 30L), marketing launch campaign 20% (BDT 24L), app development 10% (BDT 12L), working capital 10% (BDT 12L). Funding: BDT 70L angel investor, BDT 30L bank loan, BDT 20L founder savings.

END OF BUSINESS PLAN

Business Stages and Funding Options

Stage 1 – Ideation / Concept Stage

The entrepreneur develops the business idea, conducts market research, and tests feasibility.

Focus: Building a prototype or proof of concept.

Funding Source: Self-financing (Bootstrapping).

Example: A founder uses personal savings to design an app prototype.

Self-Financing (Bootstrapping)

Definition: Using personal savings, credit cards, or reinvested profits to fund the business.

Advantages: Full control, no ownership dilution, fast decisions.

Disadvantages: Limited funds, high personal risk.

Example: Michael Dell started Dell Computers with his savings.

Stage 2 – Seed Stage

At this stage, the prototype is ready and the business model is being tested.

Funds are needed for development, testing, and marketing.

Funding Sources: Friends & Family, Angel Investors.

Investors focus on the entrepreneur's potential rather than performance.

Friends and Family Funding

Definition: Raising capital from close personal networks.

Advantages: Easy access, flexible terms.

Disadvantages: Risk of harming relationships if business fails.

Example: Jeff Bezos borrowed money from his parents to start Amazon.

Angel Investors

Definition: Wealthy individuals who invest their own money in early-stage startups.

Advantages: Provide funding, mentorship, and networks.

Disadvantages: May demand equity or influence decisions.

Example: Andy Bechtolsheim invested \$100,000 in Google early on.

Stage 3 – Growth and Expansion Stage

Business starts scaling operations and entering new markets.

Funding is needed for marketing, hiring, and production.

Funding Sources: Venture Capital and Bank Loans.

Investors expect detailed financial projections and proven traction.

Venture Capitalists and Bank Loans

Venture Capitalists: Professional firms investing in high-growth startups.

Advantages: Large funding, strategic guidance.

Disadvantages: Ownership dilution.

Example: Sequoia Capital invested in Apple and Google.

Bank Loans: Traditional funding for established businesses.

Advantages: No ownership loss.

Disadvantages: Collateral and regular repayment required.

Stage 4 – Maturity / Exit Stage (IPO)

Businesses become stable and profitable, seeking large-scale growth or investor exit.

Funding Source: Initial Public Offering (IPO).

Definition: Selling shares to the public to raise capital.

Advantages: Large funds, brand visibility.

Disadvantages: Expensive, regulated, loss of control.

Example: Facebook's 2012 IPO raised \$16 billion.

- End of theory

Maths:

i) NPV, Payback Period (Thanks to Shila)

Tutorial Link:

<https://t.me/RecursionOnlineClass/411>

Examples:

Payback Period:

Solve:

Year	Cash Inflow	Cumulative
1	4000	4000
2	5000	9000
3	3000	12000
4	4000	16000

$\therefore \text{payback period} = 3 + \frac{3000}{4000}$ (राष्ट्रीय दामि)
(~~अंतर~~ राशि)
Anise should reject the project.

Q3] cut-off year is 2

Year	Cash inflow	Cumulative
1	100,000	100,000
2	60,000	160,000
3	50,000	210,000
4	60,000	270,000

$$P.P = 1 + \frac{50,000}{60,000}$$

$$= 1.83 < 2$$

∴ accept project.

can leave 22.45%

Q4] Start = 200,000

, cut-off = 2

Year	Cash inflow	Cumulative
1	10,000	10,000
2	20,000	30,000
3	30,000	60,000
4	40,000	100,000
5	30,00,000	31,00,000

$$P.P = 4 + \frac{100,000}{30,00,000}$$

$$= 4.03 \neq 2$$

reject according to P

Q5] investment = 300,000

Y	proj A	proj B	CA	CB
1	70,000	120,000	70,000	120,000
2	70,000	100,000	140,000	220,000
3	70,000	30,000	210,000	250,000
4	70,000	30,000	280,000	280,000
5	70,000	30,000	350,000	310,000

$$P_A = \frac{300,000}{70,000}$$

$$= 4.29$$

$$P_B = 4 + \frac{20,000}{30,000}$$

$$= 4.67$$

NPV:

Ques-5]

Year	Project A	Project B
0	(300,000)	(300,000)
1	80,000	120,000
2	"	100,000
3	"	70,000
4	"	60,000
5	"	50,000

$$NPV_A = 80,000 \times \frac{1 - (1+0.1)^{-5}}{0.1} - 300,000 \checkmark$$
$$= 3262 \checkmark$$

$$NPV_B = \frac{120,000}{(1.12)^1} + \frac{100,000}{(1.12)^2} + \frac{70,000}{(1.12)^3} + \frac{60,000}{(1.12)^4} + \frac{50,000}{(1.12)^5} - 300,000$$
$$= 3189 \checkmark$$

$$IRR = r_L + \left(\frac{NPV_L}{NPV_L - NPV_H} \right) \times (r_H - r_L)$$

321

best choice

Year	Proj A	Proj B
0	(800,000)	(800,000)
1	140,000	250,000
2	150,000	260,000
3	160,000	160,000
4	250,000	120,000
5	260,000	100,000

Let, lowest interest rate, $r = 15\% = 0.15$
for project A:

$$NPV_{15\%} = \frac{140,000}{1.15} + \frac{150,000}{(1.15)^2} + \frac{160,000}{(1.15)^3} + \frac{250,000}{(1.15)^4} + \frac{260,000}{(1.15)^5} - 800,000$$

$$= -187432$$

$$NPV_{6\%} = -7775$$

$$NPV_{2\%} = 98653$$

$$\therefore IRR_A = r_{2\%} + \frac{NPV_{2\%}}{NPV_{2\%} - NPV_{15\%}} \times (r_{15\%} - r_{2\%})$$

$$= 0.06 = 6.5\%$$

for Project B:

$$NPV_{15\%} = \frac{250,000}{1.15^1} + \frac{260,000}{1.15^2} + \frac{160,000}{1.15^3} - 800,000$$

$$= \frac{217,391}{1.15^4} + \frac{199,000}{1.15^5} - 800,000$$

$$= -162,480$$

$$NPV_{2\%} = 47208$$

$$i. IRR_B = r_{2\%} + \frac{NPV_{2\%}}{NPV_{2\%} - NPV_{15\%}} \times (r_{15\%} - r_{2\%})$$

$$= 0.02 + \frac{47208}{47208 - (-162480)} \times (0.15 - 0.02)$$

$$= 0.04$$

$$= 4.9\%$$

MIN

$$P.S = \frac{200,000}{1.15} + 2 = 173,913$$

Y	Proj A	P. B	P. C
0	-120,000	-180,000	-150,000
1	50,000	60,000	70,000
2	"	65,000	75,000
3	"	70,000	80,000
4	"	80,000	90,000

Capital = 300,000, $i = 10\% = 0.1$
 Calculating payback period:

Y	Proj A	Cum. A	Proj B	Cum. B	Proj C	Cum. C
1	50,000	50,000	60,000	60,000	70,000	70,000
2	50,000	100,000	65,000	125,000	75,000	145,000
3	50,000	150,000	70,000	195,000	80,000	225,000
4	50,000	200,000	80,000	275,000	90,000	315,000

$$PP_A = 2 + \frac{20,000}{50,000} = 2.4$$

$$PP_B = 2 + \frac{55,000}{70,000} = 2.79$$

$$PP_C = 2 + \frac{50,000}{80,000} = 2.06$$

$$NPV_A = 38403$$

$$NPV_B = 35497$$

$$NPV_C = 97196$$

(Proj A + Proj C)

ii) BEP (Thanks to Reya)

Tutorial Link:

<https://t.me/RecursionOnlineClass/438>

Problem 1:

Item	Value
Selling Price per unit (P)	\$150
Variable Cost per unit (VC)	\$90
Total Annual Fixed Costs (FC)	\$1,800,000
Company's Target Annual Profit (TP)	\$450,000

Questions:

1. Break-Even Point (units)
2. Break-Even Point (sales revenue)
3. Target Sales (units)

1. Break-Even Point (units)

$$\begin{aligned} \text{BEP (Units)} &= \frac{\text{Fixed Cost}}{\text{Sales Price} - \text{Variable Cost}} \\ &= \frac{1,800,000}{150 - 90} \\ &= \frac{1,800,000}{60} \\ &= 30,000 \end{aligned}$$

2. Break-Even Point (Sales Revenue)

$$\begin{aligned} \text{BEP (Sales Revenue)} &= \frac{\text{Fixed Cost}}{\text{Sales Price} - \text{Variable Cost}} * \text{Price} \\ &= 30,000 * 150 \\ &= 4,500,000 \end{aligned}$$

3. Target Sales (units)

$$\begin{aligned} \text{Target Sales(Units)} &= \frac{\text{Fixed Cost} + \text{Target Profit}}{\text{Sales Price} - \text{Variable Cost}} \\ &= \frac{1,800,000 + 450,000}{150 - 90} \\ &= 37,500 \end{aligned}$$

Problem 2:

TechNova Ltd. manufactures a premium smart fitness band. The following annual cost data and selling information are available:

Cost/Revenue Item	Amount
Factory rent	\$120,000
Components and materials per unit	\$45
Salaries of administrative staff	\$200,000
Packaging and shipping per unit	\$8
Direct labor per unit	\$27
Marketing and advertising campaigns	\$75,000
Sales commission per unit	\$10
Equipment depreciation	\$30,000
Selling price per unit	\$160

The company expects to earn an **annual profit of \$150,000**.

Required:

1. Identify which of the above costs are **fixed** and which are **variable**.
 2. Calculate the **total fixed cost** and **total variable cost per unit**.
 3. Determine the **break-even point in units** and **in sales dollars**.
 4. Compute the **number of units** that must be sold to achieve the **target profit of \$150,000**
-

1) Fixed and Variable Cost

Fixed costs

- Factory rent: \$120,000
- Salaries of administrative staff: \$200,000
- Marketing and advertising campaigns: \$75,000
- Equipment depreciation: \$30,000

Variable costs (per unit)

- Components & materials: \$45 / unit
- Packaging & shipping: \$8 / unit
- Direct labor: \$27 / unit
- Sales commission: \$10 / unit

2) Total Fixed Cost and Variable Cost Per Unit

$$\begin{aligned}\text{Total fixed cost (FC)} &= 120,000 + 200,000 + 75,000 + 30,000 \\ &= \$425,000\end{aligned}$$

$$\begin{aligned}\text{Total variable cost per unit (VC)} &= 45 + 8 + 27 + 10 \\ &= \$90 \text{ per unit}\end{aligned}$$

3) Breakeven Point in Units and Sales

Break-Even Point (units)

$$\begin{aligned}BEP(\text{Units}) &= \frac{\text{Fixed Cost}}{\text{Sales Price} - \text{Variable Cost}} \\ &= \frac{425,000}{160 - 90} \\ &= 6,072\end{aligned}$$

Break-Even Point (Sales Revenue)

$$\begin{aligned}BEP(\text{Sales Revenue}) &= \frac{\text{Fixed Cost}}{\text{Sales Price} - \text{Variable Cost}} * \text{Price} \\ &= 6,072 * 150 \\ &= 911,520\end{aligned}$$

4) Target Sales

$$\begin{aligned} \text{Target Sales(Units)} &= \frac{\text{Fixed Cost} + \text{Target Profit}}{\text{Sales Price} - \text{Variable Cost}} \\ &= \frac{450,000 + 150,000}{160 - 90} \\ &= 8,215 \end{aligned}$$

$$BEP(Q) = \frac{FC}{P - VC}$$

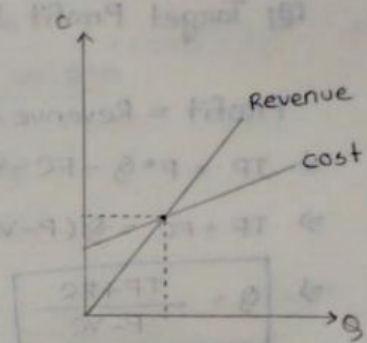
$$BEP(\$) = \frac{FC}{P - VC} \times P$$

$$= \frac{FC}{\frac{P - VC}{P}} \rightarrow \text{Contribution margin}$$

↓
यहाँ ऐसी जहाँ Fast Company Profitable है

Ex: $\frac{VC}{8}$ $\frac{P}{10}$

$$CM = \frac{P - VC}{P} = \frac{10 - 8}{10} = 0.2 = 20\%$$



BEP जहाँ Revenue बराबर Cost है
BEP ग्राम में Revenue ऐसी Cost बराबर

Math

Selling Price = 20

variable cost = 15

Fixed cost = 20000

① Find Contribution Margin

$$\rightarrow CM = \frac{P - VC}{P} = \frac{20 - 15}{20} = 0.25 = 25\%$$

② Find BEP(Q)

$$\rightarrow BEP(Q) = \frac{FC}{P - VC} = \frac{20000}{20 - 15} = 4000$$

$$\begin{aligned} \textcircled{3} \quad BEP(\$) &= BEP(Q) \times P \\ &= 4000 \times 20 \\ &= 80,000 \end{aligned}$$

$$\begin{aligned} \text{Another way, } BEP(\$) &= \frac{FC}{CM} \\ &= \frac{20,000}{0.25} \\ &= 80,000 \end{aligned}$$

iii) Ratio Analysis (Thanks to me):

Tutorial Link:

<https://t.me/RecursionOnlineClass/426>

Example Math:

	Balance Sheet		Income Statement		Ratio	
	2023	2024	2023	2024	2023	2024
Assets						
Inventories	567,678	473,625	Revenue	3,954,241	Liquidity Ratio	
Trade and other receivables	77,824	22,332	Cost of sales	(2,311,670)	Current Ratio	2.10 2.17
Advances, deposits and prepayments	114,202	26,233	Gross profit	1,642,571	Acid-Test Ratio/Quick Ratio	
Cash and cash equivalents	2,433,478	1,066,687	Operating expenses	(574,892)	Receivable Turnover Ratio	
Short term investments	1,400,000	2,896,653	Operating profit	1,067,679	Solvency Ratio	
Total current assets	4,593,182	4,485,530	Net finance income	168,306	Debt to Equity Ratio	
Property, plant and equipment	111,229	142,307	Other income	14,029	Total Debt/ Total Equity	100% 89%
Advances, deposits and prepayments	1,762	2,010	Profit before tax	1,250,014	Debt to Asset Ratio	49.95% 47.20%
Total non-current assets	112,991	144,317	Income tax expense	(225,945)	Profitability Ratio	
Total assets	4,706,173	4,629,847	Profit after tax	1,024,069	Gross Profit Ratio/ Gross Margin	41.5% 42.5%
Equity					Net Profit Ratio/ Net Profit Margin	25.9% 21.0%
Share capital	192,744	192,744			Return on Asset	15%
Reserves	5,166	5,166			Return on Equity	7%
Retained earnings	2,157,593	2,246,572			Expense Ratio	79% 86%
Total equity	2,355,503	2,444,482				
Trade payables	507,687	354,545				
Other payables and provisions	1,439,316	899,604				
Short term financing	151,955	573,820				
Advance from customers	15,041	8,282				
Lease liabilities	71,146	71,049				
Total current liabilities	2,185,145	2,067,300				
Lease liabilities	165,525	118,065				
Total non-current liabilities	165,525	118,065				
Total liabilities	2,350,670	2,185,365				
Total equity and liabilities	4,706,173	4,629,847				

All formulas:

Balance Sheet Formulas

Total Current Assets = Inventories + Trade and other receivables + Advances, deposits and prepayments + Cash and cash equivalents + Short term investments

Total Non-current Assets = Property, plant and equipment + Advances, deposits and prepayments

Total Assets = Total Current Assets + Total Non-current Assets

Total Equity = Share capital + Reserves + Retained earnings

**Total Current Liabilities = Trade payables + Other payables and provisions +
Short term financing + Advance from customers + Lease liabilities**

Total Non-current Liabilities = Lease liabilities

Total Liabilities = Total Current Liabilities + Total Non-current Liabilities

Total Equity and Liabilities = Total Equity + Total Liabilities

Income Statement Formulas

Gross Profit = Revenue – Cost of sales

Operating Expenses = Listed operating expenses amount

Operating Profit = Gross Profit – Operating Expenses

Other Income = Listed other income amount

Profit Before Tax = Operating Profit + Other Income

Income Tax Expense = Listed income tax expense amount

Profit After Tax = Profit Before Tax – Income Tax Expense

Ratio Formulas

Liquidity Ratios

Current Ratio = Current Asset / Current Liability

Acid-Test (Quick) Ratio = (Current Asset – Inventory – Prepayments or Advances) / Current Liability

Receivable Turnover Ratio = Sales (or Credit Sales) / Average Accounts Receivable

NB: If sales on credit are not given, use total Revenue as Credit Sales.

Solvency Ratios

Debt to Equity Ratio = Total Debt / Total Equity

Debt to Asset Ratio = Total Debt / Total Asset

Profitability Ratios

Gross Profit Ratio (Gross Margin) = Gross Profit / Revenue

Net Profit Ratio (Net Margin) = Net Profit / Revenue

Return on Asset (ROA) = Net Profit / Average Total Asset

Return on Equity (ROE) = Net Profit / Average Total Equity

Expense Ratio = Total Expense / Revenue

NB: Wherever "Average" is mentioned (like Average Total Asset or Average Total Equity), use the average of the two years' values:

(Average = (Value in 2023 + Value in 2024) / 2)

~ Joy